*Experiment 8: A Algorithm**

Aim

To implement the A* Search Algorithm using Python.

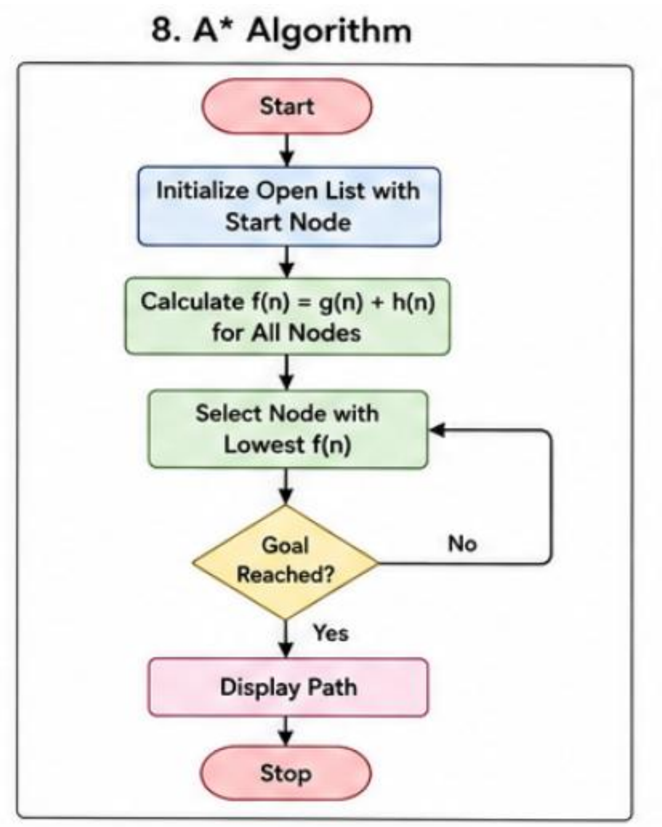
Objective

To find the shortest path between two nodes.

Algorithm

1. Select the start node.
2. Calculate f(n)=g(n)+h(n).
3. Expand the node with the lowest cost.
4. Continue until the goal is reached.
5. Display the shortest path.

Flowchart

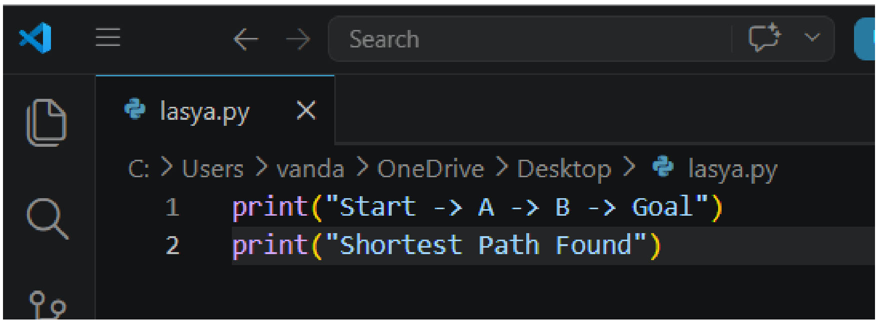


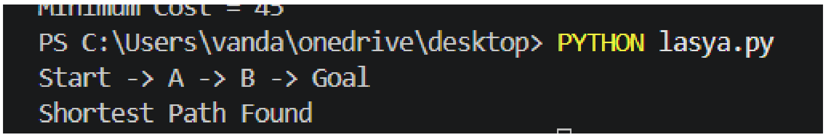
Code

```
print("Start -> A -> B -> Goal")

print("Shortest Path Found")
```

Output :





Result

The shortest path was found successfully.

Conclusion

A* efficiently finds the optimal path using cost and heuristic values.